S.Y. (B.Tech) Semester III

Subject: Statistics for Data Science

Experiment 10

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Date:	Experiment Title: Correlation and Regression Analysis
Aim	To implement hypothesis testing on Correlation and Regression Analysis
Software	Google Colab
Theory	

***Q1**. The following table gives the data on weekly family consumption expenditure(Y) and weekly family income(X)*

Y :	70	65	90	95	110	115	120	140	155	150
X :	80	100	120	140	160	180	200	220	240	260

(i) Compute the coefficient of correlation between X and Y.

(ii) Test the significance of the coefficient of correlation between X and Y at 5 percent level of significance.

Code:

i)

$X=x_i$	$Y=y_i$	x_i^2	y_i^2	x_iy_i
80	70	6400	4900	5600
100	65	10000	4225	6500
120	90	14400	8100	10800
140	95	19600	9025	13300
160	110	25600	12100	17600
180	115	32400	13225	20700
200	120	40000	14400	24000
220	140	48400	19600	30800
240	155	57600	24025	37200
260	150	67600	22500	39000
1700	1110	322000	132100	205500

$$r = \frac{n\Sigma x_iy_i - \Sigma x_i \Sigma y_i}{\sqrt{n\Sigma \bar{x}_i^2 - (\Sigma x_i)^2 (n\Sigma y_i^2 - (\bar{y}_i)^2}}$$

$$= \frac{10(205500) - (1700 \times 1110)}{\sqrt{(10 \times 322000) - (1700)^2 (10 \times 132100) - (1110)^2)}}$$

$$= \frac{168000}{\sqrt{330000 \times 88900}}$$

$$= 0.908$$

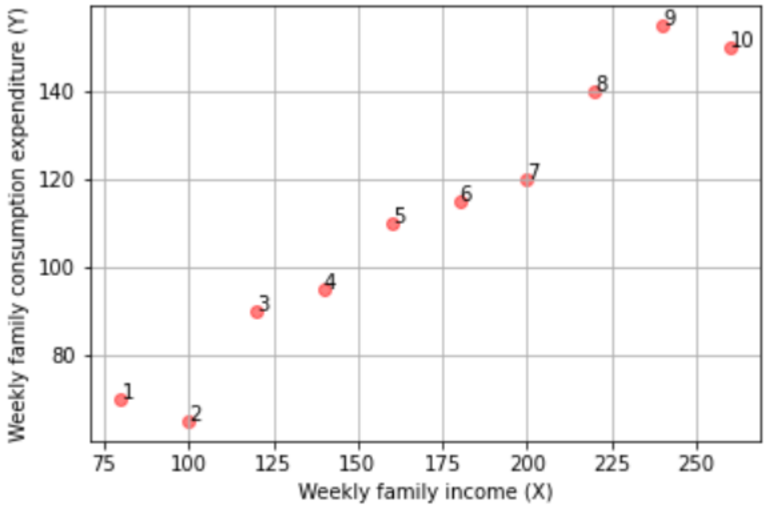
```python
import numpy as np
import pandas as pd
from scipy.stats import t
import matplotlib.pyplot as plt
import math
from statsmodels.formula.api import ols
```

```python
path = '/content/sample_data/Family_Expenditure.csv'
df = pd.read_csv(path)
df
```

	Y	X
0	70	80
1	65	100
2	90	120
3	95	140
4	110	160
5	115	180
6	120	200
7	140	220
8	155	240
9	150	260

```python
print("Scatter plot for given dataset:")
fig = plt.figure(figsize = (7,7))
X = df['X']
Y = df['Y']
n = len(X)
m = range(1, n+1)
fig, ax = plt.subplots()
ax.scatter(X ,Y , marker = 'o', c = 'red', alpha = 0.5)
plt.grid()
plt.xlabel("Weekly family income (X)")
plt.ylabel("Weekly family consumption expenditure (Y)")
for i, txt in enumerate(m):
  ax.annotate(txt, (X[i], Y[i]))
```

```
Scatter plot for given dataset:
<Figure size 504x504 with 0 Axes>
```




```python
corr_mat = np.corrcoef(X, Y)
print(corr_mat)
r = corr_mat[0,1]
print("The coefficient of correlation between X and Y is " + str(r))
```

```
[[1.         0.98084737]
 [0.98084737 1.        ]]
The coefficient of correlation between X and Y is 0.9808473685985795
```

$$t = \frac{0.9808\sqrt{8}}{\sqrt{1-(0.9808)^2}}$$

$t = 14.23$

$\alpha = 0.05$

$dof = n-2 = 8$

type = two-tailed

$t_{0.05,8} = 2.306$

$t_{cal} > t_{critical}$

∴ we reject null hypothesis

∴ There is correlation between x & y

```python
print("Null hypothesis: The coefficient of correlation between X and Y is not significant")
print("Alternate hypothesis: The coefficient of correlation between X and Y is significant")
alpha = 0.05
t_statistics = r*math.sqrt(n-2)/(math.sqrt(1-r**2))
p_value = 2*t.sf(t_statistics, n-2)
print('t_statistics = ', t_statistics, 'and p_value is ', p_value)
if p_value > alpha :
  print("Failed to reject the null hypothesis for level of significance = " +str(alpha))
else:
  print("Null hypothesis is rejected for level of significance = " +str(alpha))
```

```
Null hypothesis: The coefficient of correlation between X and Y is not significant
Alternate hypothesis: The coefficient of correlation between X and Y is significant
t_statistics =  14.243171154216448 and p_value is  5.752746116732455e-07
Null hypothesis is rejected for level of significance = 0.05
```

Q2. The following table gives the per capita household expenditure on food (Y) and per capita total household expenditure (X)

$Y:$	60	90	110	125	150	170	180	200	220	230	240	250	255	260	260
$X:$	100	150	200	250	300	350	400	450	500	550	600	650	700	750	800

(i) Compute the coefficient of correlation between X and Y.

(ii) Test the significance of the coefficient of correlation between X and Y at 5 percent level of significance.

Code:

$X = x_i$	$Y = y_i$	x_i^2	y_i^2	$x_i y_i$
100	60	10000	3600	6000
150	90	22500	8100	13500
200	110	40000	12100	22000
250	125	62500	15625	31250
300	150	90000	22500	45000
350	170	122500	28900	59500
400	180	160000	32400	72000
450	200	202500	40000	90000
500	220	250000	48400	110000
550	230	302500	52900	126500
600	240	360000	57600	144000
650	250	422500	62500	162500
700	255	490000	65025	178500

750	260	562500	67600	195000
800	260	640000	67600	208000
6750	2800	3737500	5848500	1463750

$n = 15$

$$r = \frac{n\Sigma x_i y_i - \Sigma x_i \Sigma y_i}{\sqrt{(n\Sigma x_i^2 - (\Sigma x_i)^2)(n\Sigma y_i^2 - (\Sigma y_i)^2)}}$$

$$r = \frac{15 \times 1463750 - 6750 \times 2800}{(15 \times 3737500) - (6750)^2)(15 \times 584850 - (2800)^2)}$$

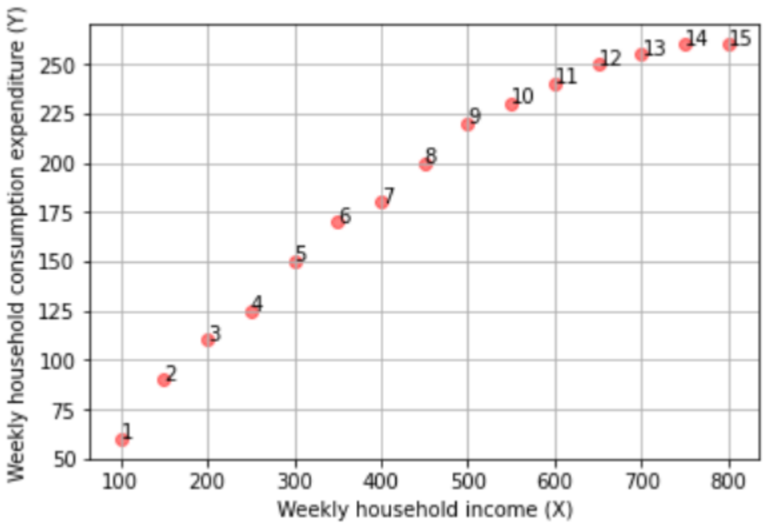
$$r = \frac{3056250}{\sqrt{10500000 \times 932750}} = 0.9766$$

```
path = '/content/sample_data/Household_Expenditure.csv'
df = pd.read_csv(path)
df
```

	Y	X
0	60	100
1	90	150
2	110	200
3	125	250
4	150	300
5	170	350
6	180	400
7	200	450
8	220	500
9	230	550
10	240	600
11	250	650

```
print("Scatter plot for given dataset:")
fig = plt.figure(figsize = (7,7))
X = df['X']
Y = df['Y']
n = len(X)
m = range(1, n+1)
fig, ax = plt.subplots()
ax.scatter(X ,Y , marker = 'o', c = 'red', alpha = 0.5)
plt.grid()
plt.xlabel("Weekly household income (X)")
plt.ylabel("Weekly household consumption expenditure (Y)")
for i, txt in enumerate(m):
  ax.annotate(txt, (X[i], Y[i]))
```

```
Scatter plot for given dataset:
<Figure size 504x504 with 0 Axes>
```




```
corr_mat = np.corrcoef(X, Y)
print(corr_mat)
r = corr_mat[0,1]
print("The coefficient of correlation between X and Y is " + str(r))
```

```
[[1.         0.97658848]
 [0.97658848 1.        ]]
The coefficient of correlation between X and Y is 0.9765884824912034
```

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

$$= \frac{0.9766\sqrt{13}}{\sqrt{1-(0.9766)^2}} = 16.37$$

$\alpha = 0.05$

$dof = n-2 = 13$

type = two-tailed

$t_{0.05,\ 13} = 2.160$

$t_{cal} > t_{critical}$

$\therefore$ We reject null hypothesi

```python
print("Null hypothesis: The coefficient of correlation between X and Y is not significant")
print("alternate hypothesis: The coefficient of correlation between X and Y is significant")
alpha = 0.05
t_statistics = r*math.sqrt(n-2)/(math.sqrt(1-r**2))
p_value = 2*t.sf(t_statistics, n-2)
print('t_statistics = ', t_statistics, 'and p_value is ', p_value)
if p_value > alpha :
  print("Failed to reject the null hypothesis for level of significance =",alpha)
else:
  print("Null hypothesis is rejected for level of significance =",alpha)
```

```
Null hypothesis: The coefficient of correlation between X and Y is not significant
alternate hypothesis: The coefficient of correlation between X and Y is significant
t_statistics =  16.368555733381395 and p_value is  4.680215495139646e-10
Null hypothesis is rejected for level of significance = 0.05
```

***Q3**. The following table gives the data on weekly family consumption expenditure(Y) and weekly family income(X)*

Y :	70	65	90	95	110	115	120	140	155	150
X :	80	100	120	140	160	180	200	220	240	260

(i) *Estimate the consumption function of the family* $Y = \beta_0 + \beta_1 X + u$

(ii) *Test the significance of the parameters at* 5 *percent level of significance.*

(iii) *Find the coefficient of determination.*

Y	70	65	90	110	95	115	120	140	155	150
X	80	100	120	160	140	180	200	220	240	26[illegible]

$n = 10$

$\mu_x = \dfrac{\Sigma x_i}{n} = 170$

$\mu_y = \dfrac{\Sigma y_i}{n} = 111$

$S_{xy} = \Sigma xy - n\mu_x\mu_y = 16800$

$S_{x^2} = \Sigma x^2 - n\mu_x\mu_y = 33000$

$\beta_1 = \dfrac{S_{xy}}{S_x^2} = 0.5091$

$\beta_0 = \mu_y - \beta_1\mu_x = 24.4545$

Coeff of determination $= \dfrac{\text{exp variation}}{\text{total variation}}$

$= \dfrac{\Sigma(\hat{y} - y_i)^2}{\Sigma(y_i - \bar{y}_i)^2}$

$= \dfrac{8552}{8890}$

$r^2 = 0.9621$

Y = B0 + B1X + U

u = y - (C - mx)

.... u = y = yhat

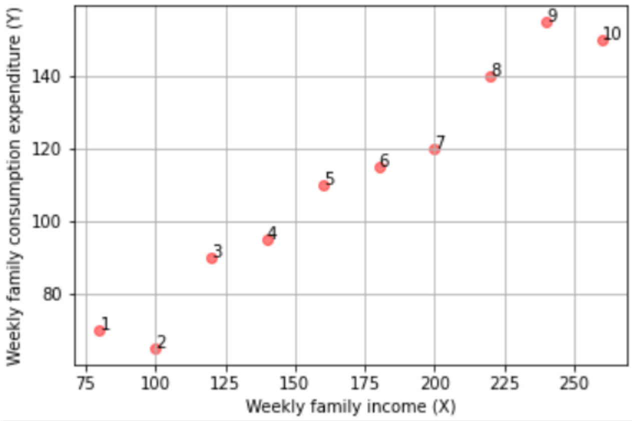
.... e = u

.... u is error

```
[ ] path = '/content/sample_data/Family_Expenditure.csv'
    df = pd.read_csv(path)
```

```
print("Scatter plot for given dataset:")
fig = plt.figure(figsize = (7,7))
X = df['X']
Y = df['Y']
n = len(X)
m = range(1, n+1)
fig, ax = plt.subplots()
ax.scatter(X ,Y , marker = 'o', c = 'red', alpha = 0.5)
plt.grid()
plt.xlabel("Weekly family income (X)")
plt.ylabel("Weekly family consumption expenditure (Y)")
for i, txt in enumerate(m):
  ax.annotate(txt, (X[i], Y[i]))
```

```
Scatter plot for given dataset:
<Figure size 504x504 with 0 Axes>
```




```
Reg1 = ols(formula = 'Y ~ X', data = df)
fit1 = Reg1.fit()
print("Regression table is given below :\n", fit1.summary())
```

```
Regression table is given below :
                            OLS Regression Results
==============================================================================
Dep. Variable:                      Y   R-squared:                       0.962
Model:                            OLS   Adj. R-squared:                  0.957
Method:                 Least Squares   F-statistic:                     202.9
Date:                Sat, 29 Jan 2022   Prob (F-statistic):           5.75e-07
Time:                        09:15:20   Log-Likelihood:                -31.781
No. Observations:                  10   AIC:                             67.56
Df Residuals:                       8   BIC:                             68.17
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept     24.4545      6.414      3.813      0.005       9.664      39.245
X              0.5091      0.036     14.243      0.000       0.427       0.592
==============================================================================
Omnibus:                        1.060   Durbin-Watson:                   2.680
Prob(Omnibus):                  0.589   Jarque-Bera (JB):                0.777
Skew:                          -0.398   Prob(JB):                        0.678
Kurtosis:                       1.891   Cond. No.                         561.
==============================================================================
```

```
Warnings:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
/usr/local/lib/python3.7/dist-packages/scipy/stats/stats.py:1535: UserWarning: kurtosistest only valid for n>=20 ... continuing anyway, n=10
  "anyway, n=%i" % int(n))
```

```python
b_0 = fit1.params['Intercept']
b_1 = fit1.params['X']
print('The estimated consumption of the family is Y_hat =',round(b_0, 4) ,' + ',round(b_1, 4), ' *X')
```

```
The estimated consumption of the family is Y_hat = 24.4545  +  0.5091  *X
```

$$Y = \beta_0 + \beta_1 x$$

$$Y = 24.4545 + 0.5091x$$

$H_0: \beta_1 = 0$, $H_a: \beta_1 \neq 0$

$$SE(\beta_1) = \sqrt{\frac{\Sigma(\hat{y} - y_i)^2}{\Sigma(x - \mu_x)^2 \times (n-2)}} = 0.0357$$

$$t = \frac{\beta_1}{SE(\beta_1)} = 14.2431$$

t critical < tstats

$\therefore$ Reject null hypothesis

$H_0: \beta_0 = 0$, $H_a: \beta_0 \neq 0$

$$SE(\beta_0) = \sqrt{\frac{\Sigma(\hat{y} - y_i)^2}{n-2} \times \left(\frac{1}{n} + \frac{\mu_x^2}{\Sigma(x - \mu_x)^2}\right)}$$

$$= 6.4138$$

$$t = \frac{\beta_0}{SE(\beta_0)} = 3.8128$$

$$t_{0.05,8} = 2.306$$

tstats > tcritical

$\therefore$ Reject null hypothesis

```
alpha = 0.05
SE_b_0 = fit1.bse['Intercept']
SE_b_1 = fit1.bse['X']
print("Hypothesis test for the significance of parameter beta_0.")
print('Null hypothesis is that beta_0 = 0.')
print('Alternate hypothesis is that beta_0 != 0.')
#t_statstics = b_0/SE_b_0
#p_value = 2*t.sf(t_statistics, n-2)
t_statistics = b_0/SE_b_0
p_value = 2*t.sf(t_statistics,n-2)
print('t_statistics = ', t_statistics, 'and p_value is ', p_value)
if p_value > alpha :
  print("Failed to reject the null hypothesis for level of significance =" ,alpha)
else:
  print("Null hypothesis is rejected for level of significance =",alpha)
print("\nHypothesis test for the significance of parameter beta_1.")
print('Null hypothesis is that beta_1 = 0.')
print('Alternate hypothesis is that beta_1 != 0.')
#t_statstics = b_1/SE_b_1
#p_value = 2*t.sf(t_statistics, n-2)
t_statistics = b_1/SE_b_1
p_value = 2*t.sf(t_statistics,n-2)
print('t_statistics = ', t_statistics, 'and p_value is ', p_value)
```

```
if p_value > alpha :
  print("Failed to reject the null hypothesis for level of significance =",alpha)
else:
  print("Null hypothesis is rejected for level of significance =",alpha)
```

```
Hypothesis test for the significance of parameter beta_0.
Null hypothesis is that beta_0 = 0.
Alternate hypothesis is that beta_0 != 0.
t_statistics =  6.2733616262066585 and p_value is  2.8632427874786518e-05
Null hypothesis is rejected for level of significance = 0.05

Hypothesis test for the significance of parameter beta_1.
Null hypothesis is that beta_1 = 0.
Alternate hypothesis is that beta_1 != 0.
t_statistics =  16.36855573338143 and p_value is  4.68021549513953e-10
Null hypothesis is rejected for level of significance = 0.05
```

```
[ ] corr_mat = np.corrcoef(X, Y)
    r = corr_mat[0,1]
    R_squared = r**2
    print("The coefficient of determination for the given regression model is",R_squared)
```

```
The coefficient of determination for the given regression model is 0.9620615604867576
```

***Q4**. The following table gives the per capita household expenditure on food (Y) and per capita total household expenditure (X)*

Y :	60	90	110	125	150	170	180	200	220	230	240	250	255	260	260
X :	100	150	200	250	300	350	400	450	500	550	600	650	700	750	800

(i) Estimate the food expenditure equation $Y = \beta_0 + \beta_1 X + u$

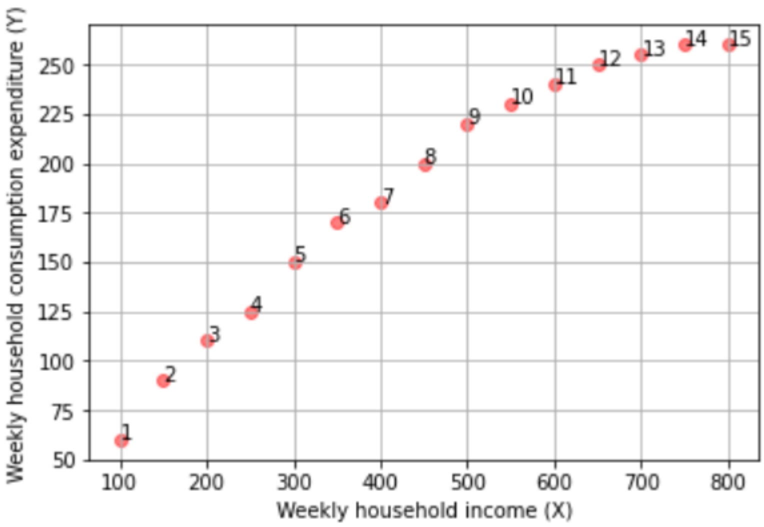
(ii) Test the significance of the parameters at 5 percent level of significance.

(iii) Find the coefficient of determination.

```
[ ] path = '/content/sample_data/Household_Expenditure.csv'
    df = pd.read_csv(path)
```

```
print("Scatter plot for given dataset:")
fig = plt.figure(figsize = (7,7))
X = df['X']
Y = df['Y']
n = len(X)
m = range(1, n+1)
fig, ax = plt.subplots()
ax.scatter(X ,Y , marker = 'o', c = 'red', alpha = 0.5)
plt.grid()
plt.xlabel("Weekly household income (X)")
plt.ylabel("Weekly household consumption expenditure (Y)")
for i, txt in enumerate(m):
  ax.annotate(txt, (X[i], Y[i]))
```

```
Scatter plot for given dataset:
<Figure size 504x504 with 0 Axes>
```



```
Reg1 = ols(formula = 'Y ~ X', data = df)
fit1 = Reg1.fit()
print("Regression table is given below :\n", fit1.summary())
```

```
Regression table is given below :
                            OLS Regression Results
==============================================================================
Dep. Variable:                      Y   R-squared:                       0.954
Model:                            OLS   Adj. R-squared:                  0.950
Method:                 Least Squares   F-statistic:                     267.9
Date:                Sat, 29 Jan 2022   Prob (F-statistic):           4.68e-10
Time:                        09:15:21   Log-Likelihood:                -60.709
No. Observations:                  15   AIC:                             125.4
Df Residuals:                      13   BIC:                             126.8
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept     55.6845      8.876      6.273      0.000      36.508      74.861
X              0.2911      0.018     16.369      0.000       0.253       0.329
==============================================================================
Omnibus:                        1.735   Durbin-Watson:                   0.328
Prob(Omnibus):                  0.420   Jarque-Bera (JB):                1.335
Skew:                          -0.666   Prob(JB):                        0.513
Kurtosis:                       2.400   Cond. No.                     1.15e+03
==============================================================================

Warnings:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.15e+03. This might indicate that there are
strong multicollinearity or other numerical problems.
/usr/local/lib/python3.7/dist-packages/scipy/stats/stats.py:1535: UserWarning: kurtosistest only valid for n>=20 ... continuing anyway, n=15
  "anyway, n=%i" % int(n))
```

```
b_0 = fit1.params['Intercept']
b_1 = fit1.params['X']
print('The estimated consumption of the family is Y_hat = ' ,round(b_0, 4) ,' + ',round(b_1, 4) , ' *X')
```

```
The estimated consumption of the family is Y_hat =  55.6845  +  0.2911  *X
```

```
alpha = 0.05
SE_b_0 = fit1.bse['Intercept']
SE_b_1 = fit1.bse['X']

print('Hypothesis test for the significance of parameter beta_0.')
print('Null Hypothesis is that beta_0 = 0')
print('Alternate Hypothesis is that beta_0 ! = 0')

t_statistics = b_0/SE_b_0
```

```python
p_value = 2*t.sf(t_statistics,n-2)

print('t_statistics = ',t_statistics, 'and p_value=',p_value)
if p_value > alpha :
  print("Failed to reject the null hypothesis for level of significance = ",alpha)
else :
  print("Null Hypothesis is rejected for level of significance =",alpha)

print('\nHypothesis test for the significance of parameter beta_1.')
print('Null Hypothesis is that beta_1 = 0')
print('Alternate Hypothesis is that beta_1 ! = 0')

t_statistics = b_1/SE_b_1

p_value = 2*t.sf(t_statistics,n-2)

print('t_statistics = ',t_statistics, 'and p_value=',p_value)
if p_value > alpha :
  print("Failed to reject the null hypothesis for level of significance = ",alpha)
else :
  print("Null Hypothesis is rejected for level of significance =" ,alpha)
```

```
Hypothesis test for the significance of parameter beta_0.
Null Hypothesis is that beta_0 = 0
Alternate Hypothesis is that beta_0 ! = 0
t_statistics =  6.2733616262066585 and p_value= 2.8632427874786518e-05
Null Hypothesis is rejected for level of significance = 0.05

Hypothesis test for the significance of parameter beta_1.
Null Hypothesis is that beta_1 = 0
Alternate Hypothesis is that beta_1 ! = 0
t_statistics =  16.36855573338143 and p_value= 4.68021549513953e-10
Null Hypothesis is rejected for level of significance = 0.05
```

```python
corr_mat = np.corrcoef(X, Y)
r = corr_mat[0,1]
R_squared = r**2
print("The coefficient of determination for the given regression model is " ,R_squared)

#if coefficient of determination (r^2) greater than 0.5 then the regression  model is good
#if coefficient of determination (r^2) less than 0.5 then the regression  model is not  good

#here coefficient of determination (r^2) is 0.95 which means the regression model is very good fit for the given dataset
```

```
The coefficient of determination for the given regression model is  0.9537250641344716
```

Conclusion	Hence we have studied and implemented Correlation and Regression Analysis.

Signature of Faculty